CCC-C-521E
March 7, 1977
SUPERSEDING
Fed. Spec. CCC-C-521D
August 17, 1966

FEDERAL SPECIFICATION

CLOTH, COATED, WINDOW SHADE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for two types of window shade cloth (see 6.1).

1.2 Classification.

1.2.1 Types. The shade cloth covered by this specification shall be of the following types (see 6.2):

Type I - Cotton cambric cloth (Translucent)
Type II - Fiberglass (Opaque)

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specification:

PPP-P-1136 - Packaging and Packing of Coated (Plastic; Rubber) and Laminated Fabrics.

Federal Standards:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).
Fed. Test Method Std. No. 191 - Textile Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specification and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

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(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 Type I - Base cloth.

3.1.1.1 Cotton cambric cloth (translucent). The base cloth shall be commercial window shade greige goods material with a minimum of 68 yarns per inch in both the warp and filling for widths up to and including 54 inches. Widths over 54 inches shall have a minimum of 56 yarns per inch in the warp direction and a minimum of 52 yarns per inch in the filling direction.

3.1.2 Type II - Base cloth.

3.1.2.1 Glass cloth. The base material shall be a 100 percent glass fiber base with a minimum yarn count of 58 total for the warp and filling directions.

3.1.3 Coating compound. The coating compound shall be polymerized vinyl chloride resin, with appropriate plasticizer and pigments to meet the requirements of this specification.

3.2 Finished shade cloth. Both sides of the base cloth (3.1.1.1) and the base material (3.1.2.1) shall be coated with the coating compound specified in 3.1.3. After coating, the cloth shall conform to the physical requirements specified in Table I when tested as specified in 4.3.2.

TABLE I. Physical Properties of finished shade cloth

Characteristic	Requirement	
	Type I	Type II
Weight ounces/sq. yd. minimum:	4.8	10.0
Breaking strength-pounds, minimum:		
All widths		
warp	50	130
filling	40	120
Tearing strength, grams, minimum:		
warp	192	1200
filling	160	600
Stiffness, centimeters, warp only:		
maximum	23.0	20.0
minimum	12.0	10.0
Flame resistance:		
After flame-seconds, maximum:		
warp	2.0	2.0
filling	2.0	2.0
Char length-inches, maximum:		
warp	5.5	4.0
filling	5.5	4.0
Color fastness, minimum:		
To light	good	good
To soap and water scrubbing	1/	1/
Resistance to cracking	2/	2/
Blocking, maximum	No. 2	No. 2

1/ The shade cloth shall show no streaks and no evidence of bleeding.
2/ There shall be no evidence of holes, breaking, or cracking.

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3.3 Opacity. (applicable only when window shade cloth is specified for blackout purposes, see 6.2). When window shade cloth is specified for blackout purposes, the diffuse luminous transmission of a single layer of the window shade cloth shall average no more than 0.001 of 1 percent; no area of the window shade cloth shall transmit more than 0.003 of 1 percent when tested in accordance with 4.4.3. The material shall have no pinholes or streaks.

3.4 Dimensions.

3.4.1 Length of rolls or pieces. The finished window shade cloth shall be furnished in rolls with a minimum length of 40 yards on a convolute or spiral wound chipboard tube. Each roll may consist of two pieces, but neither piece shall be less than 20 yards in length.

3.4.2 Width. The finished window shade cloth shall be furnished in the following overall widths in inches, as specified (see 6.2).

3.4.2.1 Type I. Cotton cambric cloth (translucent) 36, 45, 54, 72, and 77. The tolerance shall be plus or minus 1/8 inch for all widths.

3.4.2.2 Type II. Fiberglass (Opaque) 36, 45, 54, and 72. The tolerance shall be plus or minus 1/8 inch for all widths.

3.5 Identification (fiber). Each roll of window shade cloth shall have an identification ticket conforming to the requirements PPP-P-1136, except that the ticket shall be fastened to the inner or outer end of the roll with pressure sensitive tape. No pin tickets shall be used.

3.6 Workmanship. The finished window shade cloth shall conform to the quality of the product established by this specification. Cloth shall be free from cuts, tears, holes (including pinholes), scratches, abrasion marks or lines under coating resulting from imperfections of manufacture. Cloth shall also be free from uncoated, thinly coated, or thickly coated areas, and separation of coating from the base cloth. Shade cloth shall be free from any cracked, checked, or tacky areas, blisters, lumps, dirt, spot, strain or foreign matter. Edges on cloth shall show no evidence of being scalloped, curled, folded, wavy doubled or rolled. Cloth shall be free from offensive odor. Color shall be uniform, not shaded, not spotted and not mottled.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated.

4.1.2 Component testing. Each lot of base material and base cloth shall be tested in accordance with Fed. Std. 191 and the applicable test methods in Table II, to determine compliance with the requirements of section 3. Each inspection lot shall consist of a component of a single type manufactured by the same source, under essentially the same conditions and used in the production of the quantity of the end item submitted for inspection at one time. Sample shall be selected in accordance with table III, and the lot shall be unacceptable if one or more units fail to meet any requirement specified in table II.

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TABLE II

Characteristic	Specification Requirement	Test Method	No. of determinations per unit	Results reported as
Yarns per inch	3.1.1	5050	5	1 yarn
Yarns per inch	3.1.2	5050	5	1 yarn
Identification of glass fiber	3.1.2	1410	1	Pass or fail

4.2 Coating compound. The supplier shall furnish a certificate of compliance certifying that the base fabric has been properly coated and that the coating compound conforms to the requirements of 3.1.3.

4.3 Inspection of the end item.

4.3.1 Examination of the end item. The end item shall be examined for the defects set forth in the applicable subparagraphs at the inspection levels and acceptable quality levels (AQL's) in 4.3.1.4. Random samples shall be drawn from each lot for examination of visual, dimensional, and preparation for delivery defects. The lot size shall be expressed in units of rolls for examinations in 4.3.1.1 and 4.3.1.2 and in units of shipping containers in the end item inspection lot for examination in 4.3.1.3.

4.3.1.1 Examination for visual defects. The sample unit for this examination shall be one roll of shade cloth. The sample size shall be selected by taking an approximately equal number of rolls from the shipping containers sampled in accordance with 4.3.1.3. Examination for pinholes shall be in accordance with 4.4.4. Examine for the defects listed below.

Examine	Defects
Material	Any cut, tear, hole (including pinhole), scratch, or abrasion mark. Any uncoated area. Any thinly coated area. Any heavily coated area. Any separation of coating from base cloth. Any cracked or checked area. Any tacky area. Any blister or lump. Any crease or wrinkle, resulting in doubling or adhesion of surfaces. Width - less than minimum specified. Scalloped, curled, folded, wavy, doubled, or rolled edges. Any dirt, spot, stain, or foreign matter. Offensive odor.

4.3.1.2 Examination for dimensions. The sample unit shall be one roll. The sample size shall be selected from the rolls sampled in accordance with 4.3.1.1. Examine for defects listed below.

Examine	Defects
Size	Any roll (gross length) less than 40 yards. Any roll containing more than 2 pieces. Either piece less than 20 yards.

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4.3.1.3 Examination of preparation for delivery. An examination shall be made to determine whether the put up, packaging, packing and marking comply with section 5 of this specification. The examination shall be conducted as specified in PPP-P-1136.

4.3.1.4 Inspection levels and acceptable quality levels for examination. The inspection levels, for determining the sample size and the AQL's expressed in defects per 100 units, as applicable, shall be as follows:

Examination paragraph	Inspection level	AQL
4.3.1.1	S-4	4.0
4.3.1.2	S-3	4.0

4.3.2 Testing of the end item. The end item shall be tested for the applicable characteristics in Table IV. The methods of testing as specified in Fed. Std. 191, whenever applicable, and as listed in table IV shall be followed. All test reports shall contain the individual values utilized in expressing the final result. The sample unit for test purposes shall be 2 yards full width of coated cloth. The sample size (number of sample units) shall be as shown in table III. The lot shall be unacceptable if one or more units fail to meet any requirement specified. For those characteristics where test results are reported as pass or fail, there shall be no evidence of failure of any test specimen to meet the requirements as specified.

TABLE III. Sample Size

Lot size (yards)	Sample size
800 or less	2
801 and up to and including 22,000	3
22,001 and over	5

TABLE IV. Testing of the end item

Characteristic	Specification requirement	Test method	Number of determinations per unit	Results reported as: Pass or fail	Results reported as: Numerically to nearest
Weight	Table I	5041	5		0.1 ounce
Breaking strength	Table I	5102.1	5 Warp 5 Filling		Pound
Flame resistance: 1/					
After flame	Table I	5903.2	5 Warp		1/5 second
	Table I	5903.2	5 Filling		
Char length	Table I	5903T	5 Warp		0.1 inch
	Table I	5903T	5 Filling		
Colorfastness:					
Light	Table I 2/	5660.2	1		Rating
Soap and water scrubbing	Table I	para. 4.4.1	1	X	
Resistance to cracking	Table I	para. 4.4.2	1	X	
Opacity	Par. 3.3	para. 4.4.3	3		0.001 of 1 percent
Blocking	Table I	5872 3/	1		Rating
Tearing strength	Table I	5132.1	5		Gram
Stiffness	Table I	5204.1	5		1.0 millimeter

1/ The glow time shall not be reported.
2/ Except that the time of exposure shall be 200 hours.
3/ Test shall be conducted at 160°F.

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4.4 Test Methods.

4.4.1 Soap and water scrubbing. A piece of the shadecloth approximately 24 by 12 inches shall be laid on a flat horizontal surface and approximately one half gently scrubbed with a clean white cotton cloth pad that has been wet out with a lukewarm (approximately 100°F.) water solution containing 0.5 percent neutral soap. The specimen shall be scrubbed by using a unidirectional motion accomplished by allowing the pad to make contact with the surface of the cloth just after the beginning of the stroke and lifting the pad from the surface just before the end of the stroke. Each unit area of the specimen shall be scrubbed with 10 to 15 strokes at such a rate that 1 square foot shall be fully treated within 3 minutes when a 3- by 3-inch pad is used. After completion of the scrubbing with soap solution, the washed area shall be rinsed by repeating the washing operation except that lukewarm (approximately 100°F) water shall be used in place of the soap solution. The washed portion of the cloth shall be examined for streaking. Bleeding is indicated by any discoloration of the cloth during the washing or rinsing of the specimen or discoloration of the water. To determine discoloration, test shall be compared against original sample of water by viewing through test tubes.

4.4.2 Resistance to cracking. Cut a 2- by 2-inch specimen of the window shade cloth. Crease the test specimen to form a crease parallel to the direction of the filling. A five-pound weight not greater than 20k.2 inches in diameter shall be placed on the fold 10 seconds while the specimen rests on a hard surface. It shall then be folded in the same position but the opposite direction and the weight again applied. Repeat procedure until 5 double folds have been made and examine specimen for holes, cracking, or breaking. Failure is indicated if the specimen shows evidence of holes, cracking, or breaking along the crease line.

4.4.3 Opacity. (applicable only when window shade cloth is specified for blackout purposes (see 6.2). The diffuse luminous transmission of a single layer of blackout material shall be obtained by placing the window shade cloth to be tested in front of a bright diffusing source and comparing the brightness of the unlighted side of the window shade cloth with the brightness of the light source observed directly. Three specimens shall be tested from each sample unit. The specimens shall be taken from three separate areas and the results averaged. Observations shall be from the direction in which the specimen appears brightest (this is usually the perpendicular direction). The source of light shall be a diffusing screen, such as a piece of white opal glass (photometrical optical quality), with an incandescent lamp of high illumination behind it. The illumination of the test specimen shall be sufficient to permit brightness settings to be made for specimens of diffuse luminous transmission equal to 0.001 of 1 percent. The lamp shall be energized with a regulated power supply and shall operate at a color temperature of 2856K. The determinations shall be accomplished by commercial photometers such as the Pritchard Telephotometer, model 1980, manufactured by the Photo Research, 3000 N. Hollywood Way, Burbank, California 91505; the Gamma Telephotometer, Model 2009K, manufactured by Gamma Scientific Inc., 3777 Riffin Rd, San Diego, Calif. 92123 or equal.

4.4.4 Examination for pinholes. Examination for pinholes shall be made by viewing the surface of the cloth held under light tension. A bright light source (100 watt bulb) shall be positioned behind the coated cloth so as to clearly illuminate the surface without producing glare in the observer's eyes. A pinhole is defined as any opening observed in the cloth under the conditions specified above which is not visible when the coated cloth is viewed at the same distance in average daylight or the equivalent thereof.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. The cloth, put up in rolls, shall be packaged, packed and marked in accordance with PPP-P-1136. The level for put up and packaging shall be A or C, the level of packing shall be A or C and the marking shall be civil or military as specified (see 6.2).

5.1.1 Levels A and C. The cloth, put up in rolls, shall be packaged in accordance with the applicable requirements for class 5 fabrics of PPP-P-1136.

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5.2 Packing. Packing shall be level A or C as specified (see 6.2).

5.2.1 Levels A and C. The cloth, packaged as specified in 5.1, shall be packed in accordance with the applicable requirements of PPP-P-1136.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the applicable requirements of PPP-P-1136.

5.3.1 Precautionary marking. The following precautionary marking, in letters not less than three-fourths of an inch in height, shall appear at least on one side and wherever practicable on two sides of each shipping container:

KEEP COOL AND DRY

6. NOTES

6.1 Intended use. The cloth covered by this specification is intended to be used in the manufacture of window shades.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) Whether the window shade cloth is to be used "for blackout purposes" (see 3.3 and 4.4.3).
(d) Width required (see 3.4.2).
(e) Level of put up, packaging and packing required (see 5.1).
(f) Whether civil or military marking is required (see 5.1).

6.3 Potential energy savings. A Federal Energy Administration/National Bureau of Standards pamphlet entitled, "Energy Conservation with Windows" discusses the potential energy conservation benefits of shading windows during the summer against solar heat. The pamphlet lists shading devices and the percentage they can reduce solar heat gain; citing 50 percent and 44 percent for white opaque roller shades and white translucent shades respectively.

Orders for this publication are to be placed with [illegible]istration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. each.

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